FINAL PROJECT REVIEW

SonicVitals: A Low-Cost ESP32-S3 Platform for Synchronized Camera-Based Photoplethysmography and Acoustic Cardiac Sensing

Team

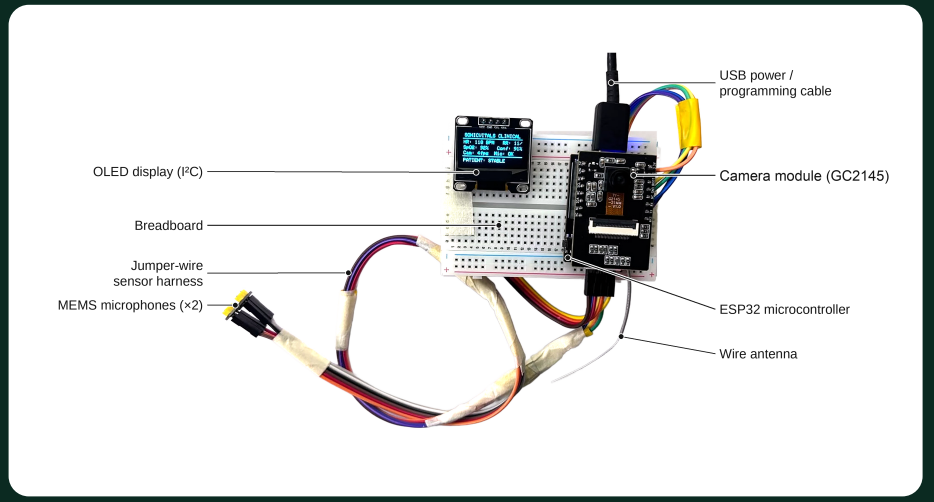
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Bench prototype: ESP32-S3 + GC2145 camera + INMP441 microphones + SSD1306 OLED

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Introduction

- Heart rate (HR) and respiratory rate (RR) are the most frequently recorded vital signs.
- They are normally measured with contact sensors: ECG electrodes, finger PPG clips, chest belts.
- Contact sensing is reliable, but uncomfortable for long recordings, unsuitable for fragile or burned skin, and a cross-infection risk.
- This motivates contactless measurement using a camera and sound.

Key idea: one cheap node captures both signals on the same clock, and the host estimates and cross-checks the vital signs.



Remote photoplethysmography (rPPG)

Each heartbeat pushes blood into skin capillaries; haemoglobin absorbs green light, so skin colour changes by 0.1–1 %. Averaging thousands of skin pixels in video recovers the pulse.



Phonocardiography (PCG)

Heart valves closing produce the S1 ("lub") and S2 ("dub") sounds at 20–150 Hz; their period gives the heart rate.



Why combine both

Camera and sound fail for different physical reasons (motion and light vs. noise), so they can check each other.

Problem statement and motivation

Problem: Monitor HR and RR without touching the person, using low-cost hardware, and know when a reading cannot be trusted.

Who needs contactless monitoring

Elderly and bedridden

People with dementia pull clips and straps off.

Fragile skin

Newborns and burn patients: adhesive sensors can injure skin.

Shared screening

Health rooms and kiosks without a shared finger clip.

What exists today

Option	Cost	Limitation
Finger pulse oximeter	₹500–1,500	Needs contact
Smartwatch / band	₹2,000+	Must be worn and charged
Face-scan phone apps	App / licence	Closed; someone holds the phone
Radar / under-mattress	₹10,000+	Single purpose, closed

Literature survey

Work	Method	Contribution	Limitation
Verkruysse et al., 2008	Green-channel rPPG, ambient light	First pulse signal from a consumer camera	Sensitive to motion and light changes
Poh et al., 2010	ICA blind source separation	Automated webcam HR	Needs a still subject
de Haan & Jeanne, 2013	CHROM chrominance projection	Cancels motion/specular component	Assumes standard skin-tone model
Wang et al., 2017	POS projection	Unified algorithmic principles of rPPG	Desktop processing; camera only
Chen & McDuff, 2018	DeepPhys CNN	Learned motion-robust pulse	Needs training data and a GPU
Villarroel et al., 2019	Video monitoring in NICU	Clinical feasibility in neonates	Costly camera setup
Liang 1997; Springer 2016	Shannon envelope; HSMM	Robust S1/S2 heart-sound segmentation	Needs a chest-contact stethoscope
Mariakakis et al., 2017	BiliScreen scleral colour	Phone-based jaundice screening	Needs a calibration accessory
McDuff 2017; Nowara 2020	Compression; skin-type meta-analysis	Compression and darker skin lower rPPG SNR	Known limits for low-cost cameras

Research gap and objectives

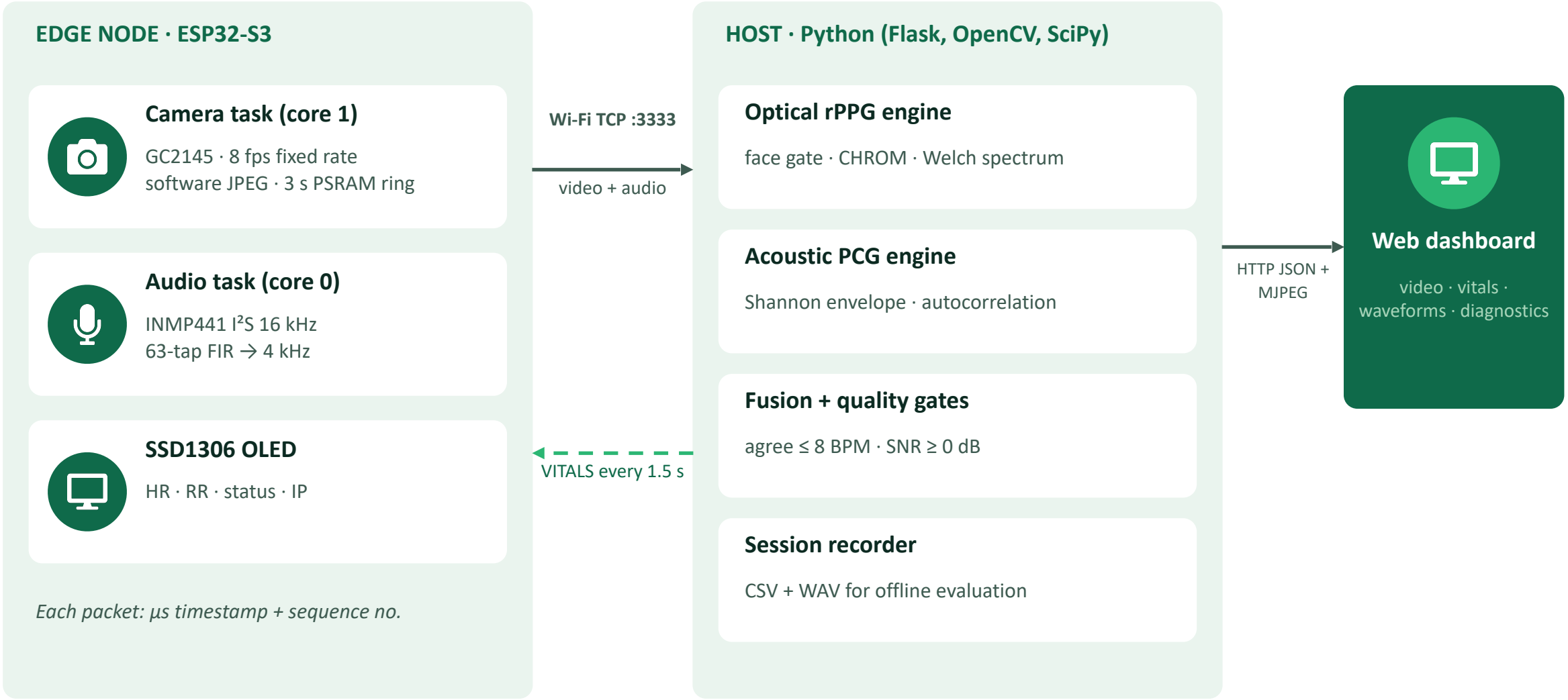
Research gap

- rPPG systems are camera-only: no independent check when motion or light corrupts the signal.
- Multimodal studies use separate devices with unsynchronised clocks, on desktop or GPU platforms.
- Few systems report when a reading is unreliable; many display a number regardless.
- Little documentation of what a very low-cost embedded camera can and cannot do.

Objectives

1. Design an ESP32-S3 node that captures video and audio with one microsecond time base.
2. Stream both reliably to a host over Wi-Fi with integrity checks.
3. Estimate HR and RR by CHROM rPPG with a face-validity gate.
4. Estimate HR from heart sounds and fuse both with an agreement gate.
5. Build a live dashboard with recording and exploratory colour screening.
6. Verify algorithms and the link quantitatively; define a validation protocol.

Proposed system architecture



Hardware components and pin mapping

Component	Key specifications	Role
ESP32-S3 (camera board)	2 × Xtensa LX7 @ 240 MHz, 512 KB SRAM, 8 MB PSRAM, Wi-Fi	Acquisition, timestamps, streaming
GC2145 camera	2 MP CMOS, DVP 8-bit, used at 240×160 RGB565	Facial video for rPPG
INMP441 microphone	I²S MEMS, 24-bit, 61 dBA SNR	Heart sounds (chest)
SSD1306 OLED	0.96", 128×64, I²C 400 kHz	Local readout of vitals and IP
Host laptop	Python 3, Flask, OpenCV, NumPy/SciPy	Signal processing, dashboard

Signal	GPIO
Camera XCLK / PCLK	15 / 13
Camera VSYNC / HREF	6 / 7
Camera SCCB SDA / SCL	4 / 5
Camera D0–D7	11, 9, 8, 10, 12, 18, 17, 16
Mic I²S SCK / WS / SD	43 / 44 / 0
OLED SDA / SCL (0x3C)	47 / 48



Pin conflict found during integration: the mic's I²S lines (GPIO43/44) are also UART0, wired to the board's only USB-serial chip (CH340). With the mic on, the USB link dies, and GPIO0 is a boot-strapping pin. Solution: stream over Wi-Fi and use USB only for power and flashing.

Firmware design

FreeRTOS tasks

Core 0 · audio task

I²S DMA read → 63-tap FIR anti-alias → decimate 16 → 4 kHz → queue (2 s)

Core 1 · camera task

Capture at fixed 8 fps → software JPEG (q 60) → 24-slot PSRAM ring (≈3 s)

Main loop

Send frames/audio, parse host commands, refresh OLED, OLED hot-plug check

Line protocol (v2)

```
<<HELLO:SonicVitals,2,wifi,mic,...>>
<<FRAME_START:seq,t_us,len>> base64 JPEG <<FRAME_END>>
<<AUDIO:seq,t_us,n,base64 PCM>>
<<STAT:fps,frames,drops,...>>
host → node: <<VITALS:HR,RR,SpO2,Q,STATUS>>
```

Engineering problems solved



No hardware JPEG on GC2145

Driver rejects JPEG mode → capture RGB565, compress in software



Auto-exposure hides the pulse

Lock exposure, gain, white balance after 3 s of settling



Hotspot throughput ≈ 70 kB/s

8 fps target, audio decimated to 4 kHz, PSRAM buffer for stalls



Uneven frame timing

Fixed-rate capture task; timestamp taken at capture, not on send



Board reset while flashing

Host never toggles DTR/RTS; Wi-Fi-only link mode

Methodology I: optical rPPG

Face gate
Haar cascade + 5 checks

→

Skin colour
forehead + cheeks mean R,G,B

→

Resample
timestamps → 15 Hz grid

→

CHROM
S = X − αY cancels motion

→

Band-pass
0.75–2.5 Hz Butterworth, zero-phase

→

HR
Welch peak, tracked; shown if SNR ≥ 0 dB

CHROM (de Haan & Jeanne, 2013)

R_n, G_n, B_n = channel ÷ its window mean

$X = 3R_n - 2G_n$

$Y = 1.5R_n + G_n - 1.5B_n$

$S = X - \alpha \cdot Y, \quad \alpha = \sigma(X) / \sigma(Y)$

$HR = 60 \cdot f_0$ (Welch peak in 0.75–2.5 Hz)

X and Y both carry the motion/specular term; subtracting αY cancels it and leaves the pulse.

Parameters and quality control

- Face gate: edge clearance, size 20–88 %, aspect 0.65–1.30, ROIs in frame, light ≥ 15 and balanced.
- Welch: 256-sample Hann segments, 50 % overlap, FFT 4096, parabolic peak interpolation.
- SNR (de Haan): power at f_0 and $2f_0$ vs. rest of 0.5–4 Hz; gate 0 dB (noise passes ≈ 5 %).
- Warm-up 10 s, buffer 30 s, update 0.5 s; gaps ≤ 2 s bridged.
- Peak tracking ±12 BPM around last accepted HR.

Methodology II: acoustic, respiration and fusion

ACOUSTIC · heart sounds

Band-pass
25–400 Hz, 4th order

→

Resample
to 1 kHz

→

Shannon envelope
$-x^2 \log x^2$, 20 ms smoothing

→

Autocorrelation
lag τ in 0.4–1.33 s

→

HR = 60/τ
shown if ρ ≥ 0.3

Respiration (camera)

Green channel band-passed 0.15–0.40 Hz (9–24 breaths/min); Welch peak f_r → RR = 60·f_r. Needs a 15 s segment and ≥ 50 % of low-frequency power at the peak, else "--".

HRV (heart sounds)

S1 peaks on the Hilbert envelope → inter-beat intervals → RMSSD = √mean(ΔIBI^2). Needs ≥ 20 clean intervals.

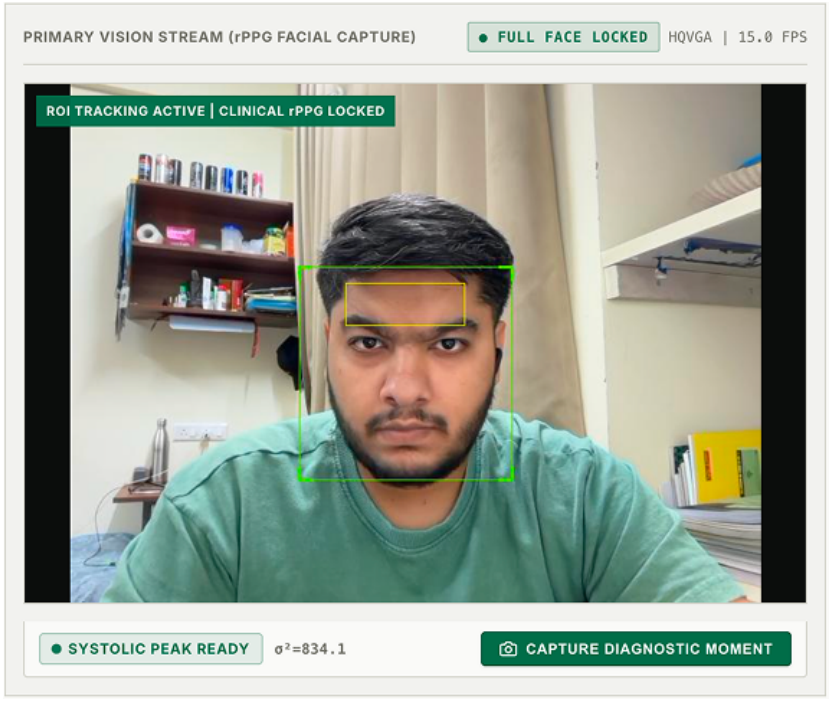
Quality-weighted fusion

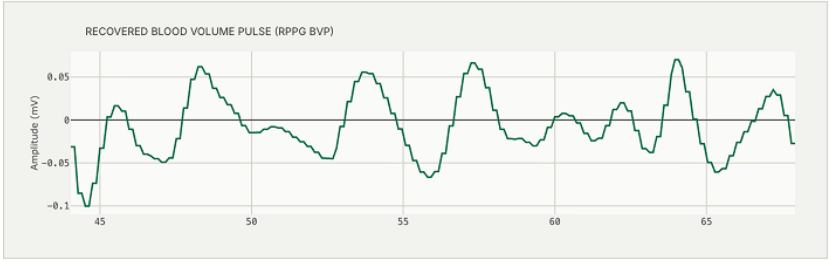
If $|HR_{cam} - HR_{snd}| \le 8$ BPM: $HR = (w_{cam} \cdot HR_{cam} + w_{snd} \cdot HR_{snd})/(w_{cam} + w_{snd})$, $w_{cam} = Q$, $w_{snd} = \rho$. Otherwise use the better channel and flag a conflict.

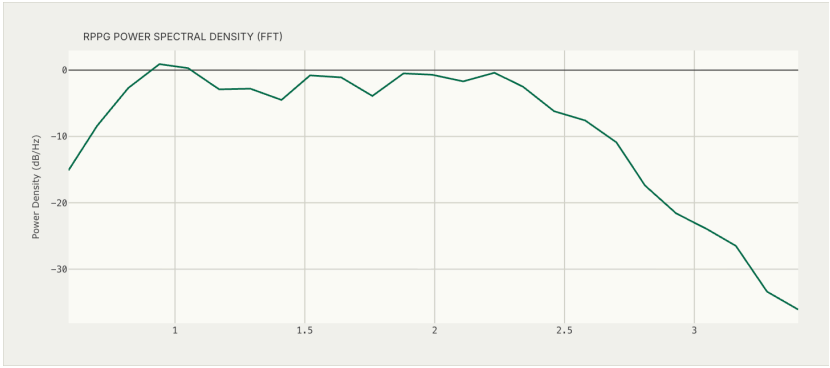
Display stabilisation

Median of last 3 fused values, then EMA; last value held ≤ 15 s while re-acquiring; "--" when no channel passes its gate.

Software implementation and dashboard







Live camera view with face gate (green) and skin ROI; recovered pulse wave (below); pulse spectrum (right).

Dashboard: live vitals, waveforms

Signal analysis: PSD, spectrogram, bands

Diagnostics: colour indices (exploratory)

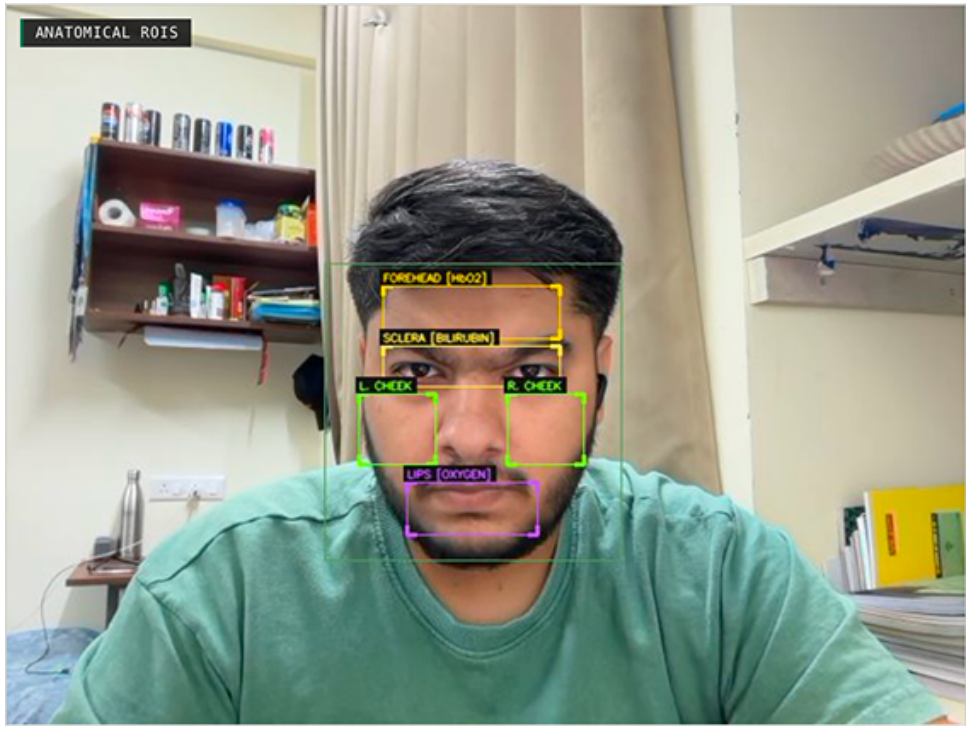
Experiments: session recording

Modules

- usb_streamer.py: link, Flask API
- rppg_engine.py: face gate, ROI
- vitals_dsp.py: shared DSP
- acoustic_engine.py: PCG
- recorder.py: sessions
- evaluate.py: offline stats
- Frontend: HTML/JS + Plotly

31 automated tests (pytest) + a node simulator

Exploratory colour screening (Diagnostics Studio)



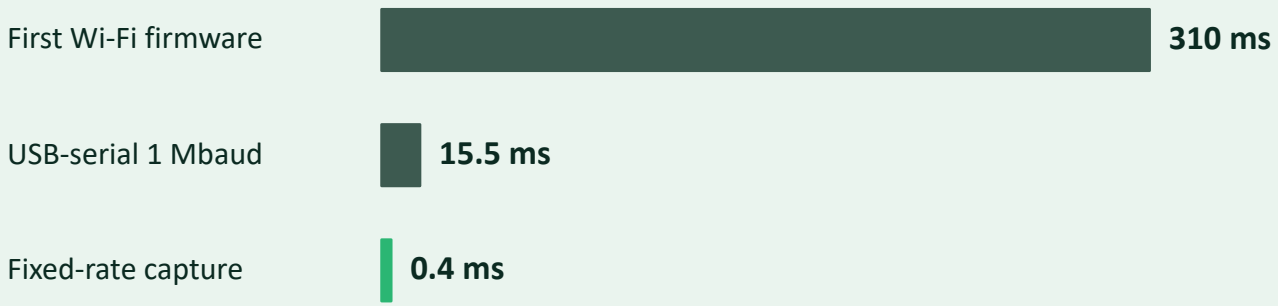
Index	ROI	Formula	Indicates
Erythema EI	Forehead	100 (log R − log G)	Perfusion
Pallor PI	Cheeks	(G + B) / 2R	Low haemoglobin
Yellowness YI	Eye band	100 (R + G − 2B) / (R + G + B)	Bilirubin (jaundice)
Cyanosis CR	Lips	B / R	Deoxygenation
Symmetry HFI	Both cheeks	100 |G_L − G_R| / (G_L + G_R)	Perfusion asymmetry
Sharpness	Whole frame	Variance of Laplacian	Frame selection

Exploratory only. Uncalibrated RGB indices depend on lighting, white balance and skin tone (e.g. warm light raises YI). Reliable use needs a colour reference card and validation against blood tests.

Results I: system performance

Configuration	Host fps	Jitter SD	Frame loss	Audio loss
Wi-Fi, first firmware (3-min session)	5.1	310 ms	0 %	0.66 %
USB-serial 1 Mbaud, mic off	7.4	15.5 ms	1.0 % rejected	mic off
Wi-Fi, fixed-rate capture (final, 30 s)	**8.0**	**0.4 ms**	**0**	**0**

Frame-timing jitter (SD)



775×

steadier frame timing

310 ms → 0.4 ms

8.0 fps

evenly spaced at host

> 5 fps Nyquist for 150 BPM

0

frames or audio blocks lost

verified by sequence numbers

Results II: algorithm verification

±2 BPM

Optical HR on synthetic signals

52, 78 and 128 BPM with irregular 6–11 fps frame timing

≈ 5 %

Noise passing the SNR gate

pure-noise test, 10–30 s windows; target ≤ 6 %

16 → 6

Buffer restarts in 3 min

replay of a recorded session after 2 s gap bridging

31 / 31

Automated tests pass

DSP, PCG, protocol, engine, server, recorder

Findings on real recordings

- First-firmware optical estimates had SNR −6 to −3 dB, i.e. at noise level, which motivated the SNR gate.
- Heart sounds through air were weak (ρ ≤ 0.18; 99.5 % of power < 25 Hz): the mic must touch the chest.

Clinical validation: planned protocol

- References: chest-strap ECG (HR, R–R), finger pulse oximeter, metronome-paced breathing.
- Conditions C1–C5: rest, talking, small motion, low light, distance.
- 30 s windows; development and test participants kept separate.
- Metrics: MAE, RMSE, Pearson r, Bland–Altman bias and 95 % limits (repeated measures).
- Acceptance: AAMI EC13, error ≤ 5 BPM or 10 %.
- Tooling ready: session recorder + analysis/evaluate.py.

HR / RR accuracy against references has not been measured yet.

Comparison with existing work

System	Contactless	Heart sounds	Synchronised single node	Open source	Quality gate shown	Validated accuracy
Finger pulse oximeter	✗	✗	—	✗	~	✓
Smartwatch / chest strap	✗	✗	—	✗	~	✓
Face-scan phone apps	✓	✗	✗	✗	~	vendor claims
rPPG research (CHROM, POS, DeepPhys)	✓	✗	✗	✓	~	on datasets
Electronic stethoscope	✗	✓	—	✗	✗	✓
SonicVitals (this work)	✓	✓	✓	✓	✓	**planned**

✓ yes ✗ no ~ partial — not applicable

Positioning: not a replacement for an oximeter, but a low-cost open platform for contactless, cross-checked monitoring.

Limitations and future work

LIMITATION	FUTURE WORK
No reference validation yet	ECG-strap study with Bland–Altman and AAMI EC13
Needs ambient light	IR LED + IR-sensitive camera for night monitoring
Motion and lighting changes	Landmark face tracking; ROI colour means on the node before JPEG
Weak heart sounds through air	Contact or stethoscope-coupled microphone
Skin tone and uncalibrated colour	Diverse participants; colour reference card
Processing on a laptop	Edge processing on the ESP32 and battery power

Conclusion

Built a low-cost ESP32-S3 node that captures camera video and heart sounds on one microsecond clock and streams them over Wi-Fi with 0 frames or audio blocks lost.

Implemented CHROM rPPG (HR, RR) and Shannon-envelope PCG (HR, RMSSD) with a five-test face gate and quality-weighted fusion with an agreement gate.

Added honest quality gating: readings that do not clearly beat noise are shown as "--".

Verified the link (8 fps, 0.4 ms jitter) and the algorithms (31 tests); the clinical validation protocol and tools are ready.

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Thank you

Questions?

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